	R V College of Engineering Department of Computer Science and Engineering(data science and cyber security) ADLDCO Retest - 1 : Question Paper

Course: (Code)CS234 AI	Applied digital logic design and computer organisation	Semester : 3rd BE

Date : Jan 2026	Duration : 120 minutes	Staff :KBR/DD/KB/PSB/MH/Sujatha Bedigar	
Name :	USN :	Section :	A/B/C/D/E/CD/CY

PART A

Sl. No	Quiz questions	Marks	* L1-L6	*CO
1.1	Solve the following using minimal number of NAND gates? (A+B)(C+D)	1	L3	CO2
1.2	Design a full adder using two half adders?	1	L2	CO2
1.3	i) Subtract the following numbers using 2's complement. Assume 8 bits 20 from 13 ii) Construct the Booth recording of a multiplier for the following sequence. 0 1 0 1 0 1 1 0 0 1 1 0 0 0 1 1 1 1 1 1 1 0	1+1 =2	L2	CO2
1.4	Design a 32 to 1 multiplexer using lower order multiplexers.	2	L2	CO2
1.5	Design a 3 to 8 line decoder using two, 2 to 4 line decoder and explain it.	2	L2	CO1
1.6	Represent the number +15.25 using 32 bit(single precision) floating point representation.	2	L2	CO1

PART B

Sl. No	Answer all the questions	Marks	* L1-L6	*CO
1a	A combinational switching network with inputs A, B, C, D should output 0 if the input combination is a valid Excess-3 coded decimal digit and 1 otherwise. Implement the network using basic gates after simplification using Kmap.	4	L2	CO2
b	Design a four bit carry look ahead adder. How does propagation delay in a Carry Look ahead Adder compare to that of parallel adder?	6	L2	CO2
2a	Realize the following function using i) 8 to 1 mux ii) 4 to 1 mux f(a,b,c,d)= Σm(0,1,2,4,5,8,9,12,13,15)	4	L3	CO2

b	Draw the flow chart for restoring division, perform the division operation on the following unsigned integers:Dividend:14 Divisor:5	6	L3	CO2
3a	Using Modified Booth's multiplication algorithm, multiply +15 with -6. (Note: +15 is the multiplicand and (-6) is the multiplier)	6	L2	CO2
b	Design a 2 bit comparator using 4 to 16 line decoder and appropriate gates.	4	L3	CO1
5	Using the Quine-McCluskey method obtain all prime implicants, essential prime implicants, and the minimized Boolean expression for: $F(A,B,C,D)=\Sigma m(4,5,8,9,12,13)+dc(0,3,7,10,11)$	10	L3	CO1
6	Illustrate the design of a single decade BCD adder(decimal adder) and show the truth table.	10	L2	CO2

Course Outcomes: After completing the course, the students will be able to:-	
CO 1	Apply design requirements for digital systems and Computer organization
CO 2	Analyze the models used for designing various Combinational and Sequential circuits
CO 3	Develop applications of synchronous sequential networks using flip flops, registers and counters
CO 4	Design optimized modern processors and memories for given specifications
CO 5	Investigate techniques of digital system design for building industry relevant real-world systems using electronic components and modern tools

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Test	Max Marks	16	44					34	26			